Xu Zichuan

Education

The University of Osaka **2025/04 – 2027/03 (Expected)**
Master of Information Science and Technology **Suita, Japan**
Huazhong University of Science and Technology **2019/09 – 2023/06**
Bachelor of Engineering in Cyberspace Security **Wuhan, China**

Work Experience

Initial S Co., Ltd. **2026/04 - Present**
Agent Development Engineer Intern | Go, React, Python, Agentic Engineering **Tokyo, Japan**

- Developed Kura-chan, a production-grade AI agent platform for NAS/edge devices, featuring a Go backend, React dashboard, and serverless AWS cloud backbone.
- Architected a unified multi-provider LLM inference layer supporting 8 backends with standardized tool-calling, and a multichannel chat system across LINE, Slack, Telegram, and WeChat via bidirectional MQTT messaging.
- Built an on-device photo intelligence pipeline using ONNX Runtime for face detection/recognition, CLIP semantic search, and perceptual hashing for deduplication—all optimized to run within NAS hardware constraints.
- Engineered 25+ agent tools (web search, shell, wiki, subagent spawning, scheduled tasks) and a dynamic skill-loading system with OpenAPI codegen, enabling extensible agent capabilities with hot-pluggable modules.

Rakuten Group, Inc. **2025/09 – 2025/10**
Applications Engineer Intern | PHP · Laravel · Python · MySQL **Tokyo, Japan**

- Developed two web applications using **PHP/Laravel**, which transform the complex and time-consuming commission exemption settings and server configuration into a real-time and automated platform.
- Automated monthly Rakuten Card validation by architecting a scalable data pipeline with **Python** and **Apache Airflow**, replacing manual, engineer-driven script executions with a zero-touch, highly reliable scheduled workflow.

Shanghai Dameng Database Co., Ltd. **2023/07 – 2025/02**
Development Engineer | Java · SQL · Linux · DBMS **Shanghai, China**

- Designed and developed the functionality of an enterprise-level database management tool, and participated in the maintenance of defects and bugs in existing database client tools. Published a **patent** about installer (CN117762435A).
- Addressed large dataset rendering bottlenecks in database management tools by refactoring the underlying SWT table drawing logic using the **NatTable** framework, slashing the rendering time for million-row query results **from minutes to seconds**. Led the architectural design and implementation of an embedded interactive terminal, deeply integrating **JediTerm** to deliver an **IDE-like terminal experience** and optimize overall usability.
- Participated in the development of the company's internal enterprise-level front-end framework DUI based on **Google Web Toolkit (GWT)**, achieving independent control over the underlying front-end technology.

NSFOCUS Technologies Group Co., Ltd. **2023/02 – 2023/04**
Research & Development Intern | Python · Django **Wuhan, China**

- Collaborated on the development of the company's security solutions, specializing in leveraging Python **Scrapy** to gather intelligence on fraudulent apps into a database and utilized Python **Django** framework for backend development of a network probing system.
- Conducted initial technological assessments of products, managed routine maintenance, resolved customer concerns, and provided technical support, including devising encryption algorithms to anonymize data for customers to use in machine learning training.

Department of Computer Science at Virginia Tech **2022/04 – 2022/09**
Remote Research Assistant | Python · PyTorch **Blacksburg, U.S.**

- Conducted research on **Graph Neural Networks (GNN)** applied to system provenance graphs; investigated graph-based intrusion detection methods to identify malicious APT (Advanced Persistent Threat) behaviors.
- Designed and implemented a novel graph representation learning framework using **PyTorch**, capable of extracting topological features from massive, sparse system audit logs for fine-grained entity classification.
- Evaluated the model on large-scale curated security datasets, achieving a state-of-the-art **97% accuracy** in predicting malicious system entities while maintaining a low false-positive rate in complex execution environments.